

1 Two Architectures for Simulating Biomarker-Treatment
2 Interactions: Implications for Statistical Power Under
3 Carryover

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44 1 Abstract

45 **Background.** Simulation studies of predictive biomarker-treatment interactions
46 in N-of-1 clinical trials require a data-generating process (DGP) that specifies how
47 the biomarker moderates treatment response. Two fundamentally different DGP
48 architectures are used in practice, and the choice between them determines the
49 extent to which carryover effects reduce statistical power to detect the interaction.
50 The published simulation literature does not distinguish them.

51 **Methods.** We formalize two architectures: direct mean moderation (Architec-
52 ture~A), under which the biomarker scales the treatment effect in the popula-
53 tion mean structure; and differential correlation (Architecture~B, the multivari-
54 ate normal approach), under which the interaction emerges from treatment-state-
55 dependent correlation between biomarker and response in the joint distribution.
56 Power consequences under both architectures are quantified by Monte Carlo sim-
57 ulation on an aggregated N-of-1 design framework with continuous-time AR(1)
58 residual correlation in the analysis model. The biomarker, treatment phase, and
59 carryover decay are parameterized to the prazosin/PTSD reference application.

60 **Results.** Under Architecture~A, power is largely preserved under carryover
61 because the proportional relationship between biomarker and drug response is
62 maintained at every timepoint. Under Architecture~B, carryover erodes the dif-
63 ferential correlation signal and reduces power. At the prazosin-calibrated refer-
64 ence cell ($N = 35$ per randomization path, i.e. effective analysis $N = 70$ for
65 the crossover (CO) and open-label-with-blinded-discontinuation (OL+BDC) de-
66 signs and 140 for Hybrid after path concatenation; $c_{bm} = 0.45$, OL+BDC de-
67 sign) across $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks, Architecture~B power drops from 0.777
68 to 0.539 (a 30.6% relative loss; $z = 7.64$, $p < 10^{-13}$ on the B-versus-A con-

trast), while Architecture~A drops only from 0.751 to 0.730 (a 2.8% relative loss). The B-versus-A absolute power-loss gap of 21.7 percentage points at OL+BDC, $t_{1/2} = 1.0$ is approximately 11 times the matched Architecture~A loss; the Hybrid design produces a smaller but still significant gap ($z = 3.96$, $p < 10^{-4}$); the traditional crossover design produces no detectable gap because its within-subject AR(1) structure dominates the carryover-mediated correlation channel that Architecture~B uses. The 36-cell production factorial (2 architectures $\times$ 3 designs $\times$ 3 $t_{1/2} \times 2 c_{bm}$ levels) was run at 1{,}000 replicates per cell with 100% convergence; the Monte Carlo standard error (MCSE) on cell power is 0.013-0.016. Type~I error sits in $[0.010, 0.074]$, conservative relative to the nominal level and with no architecture-specific inflation. The conservatism reflects the corCAR1 working-covariance miscalibration characterized in a companion calibration study¹; being common to both architectures, it does not affect the architecture contrast.

Conclusions. The two architectures encode different biological mechanisms, heterogeneity at the mean structure versus heterogeneity at the variance structure, and the choice between them is a substantive scientific question rather than a parameterization convenience. We recommend explicit declaration of the DGP architecture in any biomarker-validation simulation study and provide guidance for selecting the architecture appropriate to a given clinical context.

2 1. Introduction

[1]

The central question is biomarker-treatment interaction; simulation studies need a DGP that encodes it.

- Does a baseline biomarker predict differential treatment response, i.e., does it moderate the treatment effect?
- Closed-form power is available only in simplified cases [pmsimstats-paper08]; otherwise simulation is the principal power evaluation tool for proposed trial designs.
- These simulations require a DGP that explicitly encodes the biomarker-treatment interaction mechanism.

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A central question in precision medicine trial design is whether a baseline biomarker predicts differential treatment response, that is,

whether the treatment effect is moderated by the biomarker value. For simplified special cases, such as a balanced strict repeated-measures ANOVA under sphericity, the power of the interaction test admits a closed-form expression, derived together with its equivalence to a two-sample t -test in a companion paper². For the continuous-time mixed model and the N-of-1 designs considered here, however, no such closed form is available, so simulation studies are the principal tool for evaluating the statistical power of proposed trial designs to detect such interactions, and these simulations necessarily require a data-generating process (DGP) that encodes the biomarker-treatment interaction mechanism.

[2]

Every simulation framework must choose how to generate treatment effect heterogeneity, and that choice has consequences.

- Heterogeneity of treatment effects (HTE) literature (2005, 2020) distinguishes qualitative interactions (direction reverses) from quantitative ones (magnitude varies).
- Any simulation framework must decide how to encode this heterogeneity in the DGP.
- That decision is substantive, not a mere parameterization detail.

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The statistical literature on treatment effect heterogeneity, developed notably by Rothwell³ and synthesized more recently by Kent et al.⁴, distinguishes between *qualitative* interactions, in which the direction of the treatment effect reverses across biomarker strata, and *quantitative* interactions, in which the magnitude varies but the direction remains consistent. As we shall argue, any simulation framework intended to evaluate power for biomarker-moderated effects must choose how to generate this heterogeneity, and that choice is not merely a technical convenience.

[3]

Architecture A (direct mean moderation) is the near-universal standard; the Hendrickson (2020) multivariate normal (MVN) approach is apparently unique.

- Every major simulation framework uses direct mean moderation: Simon (2010) for stratified designs; Freidlin and Korn

- (2014) for adaptive enrichment; Renfro and Sargent (2017) for basket/umbrella trials; all N-of-1 frameworks.
- Wang and Schork (2019) and Schork (2022) also use Architecture A despite sharing authorship with Hendrickson et al. (2020), suggesting the MVN parameterization is a paper-specific choice, not a lab convention.
 - To the best of our knowledge, no other clinical trial simulation framework uses differential correlation (MVN) to encode the biomarker-treatment interaction.

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The dominant approach in the clinical trial simulation literature is direct mean moderation, in which the biomarker enters the outcome model as an interaction term that explicitly scales the treatment effect. Simon⁵ employs this approach throughout the simulation frameworks he develops for biomarker-stratified designs; Freidlin and Korn⁶ adopt it for adaptive enrichment trials; Renfro and Sargent⁷ use it throughout their treatment of basket and umbrella trials; and it underlies the platform trial literature without exception. Recent simulation work on biomarker-treatment interaction estimation in randomized trials with prognostic covariates, specifically the study by Haller and Ulm⁸, likewise adopts the standard regression specification with explicit interaction terms. Among N-of-1 simulation studies, Zucker et al.⁹, Araujo et al.¹⁰, and Duan et al. (2013) similarly use hierarchical random-effects models in which the biomarker predicts the individual treatment effect through the mean structure. Notably, Wang and Schork¹¹ and Schork¹², whose methodological work addresses power and design for N-of-1 trials with serial correlation and carryover directly, also specify the treatment effect through a linear mean structure, which suggests that the multivariate-normal differential-correlation parameterization of Hendrickson et al.¹³ is not a convention of any particular research program but rather a paper-specific design choice. To the best of our knowledge, no other clinical trial simulation framework generates the biomarker-treatment interaction through differential correlation in a multivariate normal joint distribution rather than through direct mean moderation.

[4]

Architecture B is principled, not merely idiosyncratic: it is the reduced-form second-moment encoding of a latent responder-class mechanism.

- If the biomarker is a proxy for an unobserved responder subtype, the faithful DGP is a finite mixture; a single MVN reproduces its first two moments.
- Architecture B is thus one point on a continuum from continuous-heterogeneity to explicit latent-class formulations.
- When to prefer it, and whether the approximation is adequate, is taken up in Section 4.2 and formally in a companion paper [pmsimstats-paper03].

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Although the differential-correlation parameterization is unusual in the simulation literature, it is not arbitrary. It is the reduced-form, second-moment representation of a latent responder-class mechanism: if the biomarker does not itself determine the drug’s effect but instead indexes an unobserved subpopulation that responds (a disease subtype, an inflammatory state, a genetic background), the biologically faithful DGP is a finite mixture, and a single multivariate normal reproduces that mixture’s first two moments through a treatment-state-dependent biomarker-response correlation. On this reading Architecture~B is not a competitor to mean moderation on equal footing but the covariance-only end of a continuum that runs through mean-and-covariance and location-scale specifications to an explicit latent-class model. When this reduced form is preferable to direct mean moderation, and when the approximation breaks down, is taken up in Section~4.2 and is the subject of a companion paper¹⁴ that develops this biomarker-moderation spectrum and evaluates class-aware analyses at trial-relevant sample sizes.

In plain language, and without the latent-variable vocabulary: suppose patients come in two kinds, those who respond to the drug and those who do not, but we cannot see which kind any given patient is. The biomarker is only an imperfect clue to that hidden status, so two patients with the same biomarker value can still respond differently. Architecture~A cannot capture this, because it assumes the biomarker fixes the size of each patient’s drug effect exactly. Architecture~B captures it by letting the biomarker and the response track each other only while the drug is acting: a higher biomarker tends to go with a larger response, as a statistical tendency rather than a fixed rule. When the drug washes out that tendency fades, which is the reason carryover costs more power under Architecture~B than under Architecture~A.

The comparison originates not in an error but in recognizing that Hendrickson’s Architecture B is appropriate for its latent-subtype biomarker, whereas the N-of-1 simulation literature defaults to Architecture A.

- Hendrickson et al. (2020) use Architecture B, the correct encoding for a biomarker hypothesized to proxy a latent responder subtype (resting systolic blood pressure for a noradrenergic PTSD subtype).
- Most N-of-1 and biomarker-trial simulation work instead encodes the interaction through Architecture A.
- The divergence became concrete when two implementations of the same pmsimstats pipeline, one using B and one A, reached different conclusions about carryover.

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Architecture~B is the appropriate data-generating choice for the trial that motivated Hendrickson et al.¹³. The predictive biomarker there is resting systolic blood pressure¹⁵, which is hypothesized to index a latent noradrenergic subtype of PTSD preferentially responsive to alpha-1 adrenergic blockade¹⁶ rather than to determine prazosin exposure. On that hypothesis the biomarker is a proxy for an unobserved responder class, exactly the latent responder-class mechanism for which the differential-correlation encoding is the reduced form (Section~4.3). What brought the architectural distinction into focus was the realization that this choice is not shared by the wider N-of-1 and biomarker-trial simulation literature, which, as reviewed above, encodes the interaction almost exclusively through direct mean moderation (Architecture~A). The mismatch became concrete during development of the pmsimstats framework: the original Hendrickson-derived code generated data under Architecture~B, while a parallel reimplementaion adopted Architecture~A, and the two produced qualitatively different conclusions about how carryover erodes power. The distinction is therefore not a coding artifact to be corrected but a substantive modeling choice. The same biomarker-moderation question, encoded in the first versus the second moment of the outcome distribution, yields materially different power under carryover, and the encoding that is appropriate depends on the biology of the biomarker.

281 **This paper is the first systematic comparison of the two DGP**
282 **architectures; we formalize, simulate, and give guidance.**

- 283 • We formalize the distinction between Architecture A (mean mod-
284 eration) and Architecture B (differential correlation).
- 285 • We demonstrate power consequences through Monte Carlo simu-
286 lation.
- 287 • We offer guidance for investigators on how to choose the appro-
288 priate architecture.

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295 We present here what we believe to be the first systematic comparison
296 of these two DGP architectures for biomarker-treatment interaction
297 power estimation under carryover. We shall formalize the distinction
298 between them, demonstrate its consequences through Monte Carlo
299 simulation, and offer guidance for investigators designing simulation
300 studies for predictive biomarker-moderated treatment effects.

301 **Related documents in this series.** The mathematical derivation,
302 code-level implementation details, four implementation attempts (three
303 failed and one correct), and the empirical evaluation of the original
304 Hendrickson¹³ parameter grid under Architecture~A are documented in
305 `19-mean-moderation-implementation-notes.tex` (with a brief markdown
306 summary in `19-mean-moderation-implementation-notes.md`). A condensed
307 empirical-results summary suitable for collaborator presentation is provided in
308 `21-architecture-comparison-summary.md`.

309

310 **3 2. Two DGP Architectures**

311 **Notation.** Throughout this paper, B_i denotes the biomarker value of participant
312 i , D_{it} the binary on-drug indicator at timepoint t (1 on drug, 0 off drug), and BR_{it}
313 the biological response component of the outcome. These three symbols describe
314 the data-generating-process layer. The analysis-model layer uses a related but
315 distinct continuous drug exposure variable, Dbc , defined in Section 3.1. Code-
316 style identifiers in typewriter font (`bm`, `Dbc`, `BR`) refer to specific variables in the
317 `pmsimstats` codebase or to R formula syntax; mathematical symbols are used
318 otherwise.

3.1 2.1 Architecture A: Direct Mean Moderation

In this approach the biomarker enters the mean structure of the outcome directly. The treatment effect for participant i at timepoint t is:

$$Y_{it} = \mu_0 + \beta_1 \cdot D_{it} \cdot (1 + \beta_{bm} \cdot B_i) + \epsilon_{it}$$

where D_{it} is the binary on-drug indicator (or a continuous measure of drug exposure), B_i is the participant's biomarker value (typically centered), and β_{bm} is the biomarker moderation parameter. The interaction is explicit in the mean: participants with higher biomarker values have proportionally larger treatment effects.

Key properties:

- The interaction is **deterministic** given the biomarker value and treatment status.
- The biomarker moderation β_{bm} is a **population-level parameter** that applies uniformly to all participants.
- The signal for the interaction is in the **first moment** (conditional mean) of the outcome distribution.
- Adding noise, random effects, or carryover to D_{it} scales the entire treatment effect (including its biomarker-dependent component) proportionally. The **ratio** of drug effect between high-biomarker and low-biomarker participants is preserved.

[7]

The multiplicative and additive forms of Architecture A are interchangeable for power calculations.

- The multiplicative form maps directly to the standard additive regression, the effect-modeling equation of Kent et al. (2020).
- $\beta_3 = \beta_1 \cdot \beta_{bm}$; the additive form additionally allows a prognostic main effect.
- Both forms fall within Architecture A and are interchangeable for power calculations on the interaction coefficient.

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We note that the multiplicative form above maps directly onto the standard additive regression form for predictive biomarker effects used in the trial methodology literature, specifically the effect modeling

equation of Kent et al.⁴:

$$Y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 B_i + \beta_3 D_{it} B_i + \epsilon_{it}$$

The interaction coefficient β_3 in the additive form corresponds to the product $\beta_1 \cdot \beta_{bm}$ in the multiplicative form. The additive form additionally accommodates a prognostic main effect β_2 for the biomarker, which the multiplicative form sets to zero by construction. For the purpose of power calculations on the interaction coefficient, the two forms are interchangeable up to this prognostic-effect distinction, and both fall within Architecture A.

This architecture is used in the pedagogical simulation (`implementations/simple/simulation.R`) and in the exploratory carryover analyses (`simulation_carryover_spectrum.R`) of the pmsimstats project.

3.2 2.2 Architecture B: Differential Correlation (MVN)

In this approach the biomarker and the biological response (BR) component are jointly drawn from a multivariate normal distribution with treatment-state-dependent correlation:

$$\begin{pmatrix} B_i \\ BR_{it} \end{pmatrix} \sim \text{MVN} \left(\begin{pmatrix} \mu_B \\ \mu_{BR}(t, D_{it}) \end{pmatrix}, \Sigma(D_{it}) \right)$$

where the covariance matrix Σ has the following treatment-state-dependent BM-BR correlation:

$$\text{Cor}(B_i, BR_{it}) = \begin{cases} c_{bm} & \text{if } D_{it} = 1 \text{ (on drug)} \\ c_{bm} \cdot e^{-\lambda \cdot t_{sd}} & \text{if } D_{it} = 0 \text{ (off drug)} \end{cases}$$

where t_{sd} is time since drug discontinuation. The exponential-decay form is the default; the limiting cases $\lambda = 0$ (no decay; correlation persists indefinitely during the off-drug phase) and $\lambda \rightarrow \infty$ (immediate decay; correlation drops to zero the instant the drug is discontinued) recover the two no-carryover model variants.

[8]

In Architecture B the interaction is in the covariance structure, not the mean; it emerges from the conditional distribution given the biomarker value.

- The population mean of BR_{it} is the same for all participants; no mean-structure interaction exists.
- Given biomarker value b , the conditional expectation of BR scales with c_{bm} , creating an interaction-like structure.

- Correlation is c_{bm} on drug and decays as $c_{bm}e^{-\lambda t_{sd}}$ off drug; moderation attenuates during the off-drug period.

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The interaction is not in the mean structure: the population mean of BR_{it} is the same for all participants with the same treatment history. Instead it emerges from the **conditional distribution**: given a participant's biomarker value $B_i = b$, the conditional expectation of their biological response is:

$$E[BR_{it}|B_i = b, D_{it} = 1] = \mu_{BR}(t) + c_{bm} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b - \mu_B)$$

This conditional expectation has an interaction-like structure: the biomarker value b modulates the expected BR. The effective correlation is c_{bm} on drug and $c_{bm} \cdot e^{-\lambda t_{sd}}$ during the off-drug period; the moderation therefore persists with attenuated strength during off-drug periods and vanishes only in the limit of complete decay.

Key properties:

- The interaction is **probabilistic**: it manifests as a tendency, not a deterministic scaling.
- The biomarker moderation c_{bm} is a **correlation parameter** that governs the strength of the association.
- The signal for the interaction is in the **second moment** (covariance structure) of the joint distribution.
- Carryover effects in the DGP inflate the BR means during off-drug periods, making them resemble on-drug means. This reduces the **differential** in the correlation structure between treatment states, weakening the signal that the analysis model detects.

This architecture is used in the Hendrickson et al.¹³ publication code and the audited R/ package in pmsimstats-ng.

A third possibility is that both channels operate at once: the biomarker could scale the mean drug effect (Architecture~A) while simultaneously altering the biomarker-response correlation during active treatment (Architecture~B). Such a combined architecture, carrying independent weights on the two channels, recovers A and B as its single-channel limits and is the natural framework when the operative mechanism is uncertain. Because it introduces a second free parameter and its power behavior warrants separate treatment, we develop it in a companion

paper¹⁷ and restrict the present paper to the A-versus-B contrast.

3.3 2.3 Mathematical Comparison

[9]

Under Architecture A, carryover shrinks the signal amplitude but preserves the ratio between biomarker groups, so identifiability is maintained.

- The expected outcome difference between two participants scales linearly with D .
- Under carryover, D is replaced by a decayed value and the absolute magnitude shrinks.
- But the ratio of drug effect between high- and low-biomarker participants is preserved at every exposure level; the signal contracts but remains identifiable.

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We begin the comparison by considering the expected outcome difference between two participants with biomarker values b_1 and b_2 at the same drug exposure level D under Architecture A:

$$E[Y \mid b_1, D] - E[Y \mid b_2, D] = \beta_1 \cdot \beta_{bm} \cdot (b_1 - b_2) \cdot D$$

The absolute magnitude of this difference scales linearly with D and therefore shrinks during off-drug periods when D is replaced by an exposure-decayed value $D \cdot \text{carryover}$. We note, however, that the **ratio** of the drug effect between high-biomarker and low-biomarker participants is preserved at every level of exposure. The biomarker-treatment signal contracts in amplitude but does not lose its identifiability under carryover.

[10]

Under Architecture B, the biomarker-dependent component of the off-drug response decays exponentially, eventually eliminating the differential signal.

- Off-drug, the conditional expectation includes a biomarker-dependent term scaled by $c_{bm}e^{-\lambda t_{sd}}$.
- As t_{sd} grows, that term shrinks toward zero.

- In the limit, all participants converge to the same off-drug mean regardless of biomarker value; the interaction signal vanishes.

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Under Architecture B, the analogous quantity is:

$$E[BR|b_1, D = 1] - E[BR|b_2, D = 1] = c_{bm} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b_1 - b_2)$$

During off-drug periods with carryover:

$$E[BR|b, D = 0, t_{sd}] = \mu_{BR, \text{off}}(t_{sd}) + c_{bm} \cdot e^{-\lambda t_{sd}} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b - \mu_B)$$

Here the biomarker-dependent component decays exponentially with time off drug. As t_{sd} increases, the off-drug conditional expectation converges to $\mu_{BR, \text{off}}$ for all biomarker values, eliminating the differential signal entirely.

4 3. Consequences for Statistical Power Under Carryover

4.1 3.1 Empirical demonstration

[11]

The two DGPs are identical except for how they encode the BM-BR linkage; any power divergence is attributable solely to that difference.

- All non-biologic components (time-varying, placebo-belief, residual noise) are generated identically in both architectures.
- The only difference: Architecture B encodes BM-BR interaction in Σ ; Architecture A adds a post-draw mean shift to on-drug BR.
- The same `nlme::lme` analysis model with `corCAR1` is used for both, so divergence in results is attributable to the DGP alone.

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493 We ran identical simulation configurations under both architectures using
494 the pmsimstats-ng framework, matching the DGP parameters as
495 closely as possible between the two approaches. We note that the non-
496 biologic response components (time-varying response, placebo-belief
497 accumulation, and residual noise) are generated from the same mul-
498 tivariate normal draw with identical means, variances, within-factor
499 AR(1) autocorrelation, and cross-factor correlations in both architec-
500 tures. The two DGPs differ *only* in how the biomarker-to-biological-
501 response (BM-BR) linkage is encoded: Architecture~B embeds it in
502 the off-diagonal c_{bm} entries of Σ , whereas Architecture~A omits those
503 entries and adds a post-draw shift $c_{bm} \cdot B_i^* \cdot \sigma_{BR}$ to on-drug BR columns
504 (see 19-mean-moderation-implementation-notes.tex for the scal-
505 ing derivation). The divergent carryover behavior reported below is
506 therefore attributable to the BM-BR parameterization, not to any dif-
507 ference in how the placebo, time-varying, or residual-noise components
508 are generated. The analysis model was the same in both cases: an
509 `nlme::lme` fit with a `corCAR1` continuous-time autocorrelation struc-
510 ture and a single biomarker-by-drug interaction term, written `bm:Dbc`
511 in R formula syntax.

512 [12]

513 **Dbc is a continuous drug exposure variable that captures ex-**
514 **ponential off-drug decay; the `bm:Dbc` interaction tests whether**
515 **biomarker moderates that exposure-decayed effect.**

- 516 • `Dbc` = 1 on drug; = $\exp(-\lambda t_{sd})$ off drug, matching the DGP
517 decay rate.
- 518 • Unlike the binary D_{it} , `Dbc` smoothly captures off-drug shrinkage.
- 519 • The test of `bm:Dbc` is the test of whether the biomarker moderates
520 the (decay-adjusted) drug effect.

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527 The analysis-model drug exposure variable `Dbc` is the *continuous* ana-
528 logue of the binary DGP indicator D_{it} . It equals 1 during on-drug
529 periods and $\exp(-\lambda t_{sd})$ during off-drug periods, where t_{sd} is time
530 since drug discontinuation and λ is the analysis-model carryover de-

cay rate (typically derived from the drug's pharmacokinetic half-life as $\lambda = \ln 2/t_{1/2}$, in which case it matches the DGP decay rate of Section 2.2 by construction). Whereas D_{it} flips abruptly between 0 and 1, Dbc captures the smooth shrinkage of residual drug effect during the off-drug period. The interaction coefficient on `bm:Dbc` is therefore the test of whether the biomarker moderates the exposure-decayed drug effect.

We follow the ADEMP framework of Morris, White, and Crowther (2019): Aims, Data-generating mechanisms, Estimands, Methods, Performance measures.

Setup:

- Trial designs: CO (traditional crossover), Hybrid (N-of-1 Hybrid), OL+BDC (open-label with blinded discontinuation)
- Sample size: $N = 35$ per randomization path (effective analysis $N = 70$ for CO and OL+BDC, 140 for Hybrid, after the paths are concatenated)
- Biomarker correlation / moderation: $c_{bm} = 0.45$ / $\beta_{bm} = 0.45$
- DGP carryover half-life: $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks
- Analysis: matched Dbc with same half-life
- Replicates: 1,000 per cell (MCSE on a binomial power estimate near 0.5 is approximately $\sqrt{0.5 \cdot 0.5/1,000} \approx 0.016$)
- Implementation: 8-core `parallel::mclapply`; full factorial 2 architectures $\times$ 3 designs $\times$ 3 $t_{1/2}$ $\times$ 2 c_{bm} levels = 36 cells, 36,000 fits, 100% convergence

[13]

$N = 35$ was chosen because it keeps power off the ceiling, making any carryover-induced loss visible; Type I error is in the expected range.

- $N = 35$ is the smallest prazosin-PTSD-calibrated sample size in the literature.
- Neither architecture saturates power at zero carryover, so losses are mechanically observable.
- Type I error across 18 null cells sat in $[0.010, 0.074]$, consistent with the conservative-to-nominal range expected for this specification.

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The $N = 35$ choice is the smallest of the prazosin-PTSD- calibrated sample sizes used in the published methodological literature; it is a regime in which neither architecture saturates power at the no-carryover cells, so any carryover-induced erosion is mechanically visible. Type I error under the $c_{bm} = 0$ null cells sat in $[0.010, 0.074]$ across the 18 null cells. These rates below the nominal 0.05 are not architecture-specific; they reflect a known conservatism of the standard `corCAR1` Wald test, whose model-based standard error is overstated by roughly 6 to 10 percent under this working-covariance specification. A companion calibration study¹ localizes this miscalibration to the `corCAR1` residual layer and shows that a CR2 bias-reduced cluster-robust standard error restores the realized Type I error to approximately the nominal level, with a near-uniform power increment that leaves the ranking of analysis specifications unchanged. Because the conservatism is common to both architectures, it is common-mode with respect to the B-versus-A contrast that is the object of this section and does not affect it.

Results under Architecture A (direct mean moderation):

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
CO	0.739	0.739	0.725
Hybrid	0.992	0.987	0.978
OL+BDC	0.751	0.748	0.730

[14]

Architecture A shows modest power loss under carryover (1–3%), below preliminary expectations from earlier exploratory runs in this project.

- Relative losses: 1.9% (CO), 1.4% (Hybrid), 2.8% (OL+BDC).
- All three fall below the 8–13% range suggested by earlier exploratory runs in this project.
- Qualitative finding: modest erosion; power essentially intact under carryover.

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Power declines modestly under Architecture A: relative loss from $t_{1/2} = 0$ to $t_{1/2} = 1$ is 1.9% (CO), 1.4% (Hybrid), and 2.8% (OL+BDC). We note that all three values fall well below the 8-13% relative-loss range suggested by earlier exploratory runs in this project for direct-mean-moderation DGPs, though they reproduce

the qualitative finding of those runs, namely a modest erosion that leaves power essentially intact.

Results under Architecture B (differential correlation, MVN):

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
CO	0.745	0.754	0.723
Hybrid	0.995	0.980	0.944
OL+BDC	0.777	0.694	0.539

[15]

Architecture B shows sharply larger power loss under carry-over, especially OL+BDC (30.6%); the B-vs-A gap is significant for OL+BDC and Hybrid but not CO.

- Relative losses: 3.0% (CO), 5.1% (Hybrid), 30.6% (OL+BDC); the OL+BDC loss is $\approx 11\times$ the matched Architecture A loss.
- B-vs-A gap: $z = 7.64$ ($p < 10^{-13}$) at OL+BDC; $z = 3.96$ ($p < 10^{-4}$) at Hybrid.
- CO gap not significant ($z = 0.29$): the CO design's within-subject AR(1) structure dominates the correlation-decay channel that Architecture B depends on.

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Power declines more sharply under Architecture B, and most markedly so in the OL+BDC design: relative loss from $t_{1/2} = 0$ to $t_{1/2} = 1$ is 3.0% (CO; not significantly different from Architecture A's CO loss), 5.1% (Hybrid), and **30.6% (OL+BDC)**. The CO cell shows a small, non-monotone fluctuation ($0.745 \rightarrow 0.754 \rightarrow 0.723$), which is within one MCSE and is consistent with the null finding of no significant architecture-by-carryover interaction in that design. The OL+BDC loss is approximately 11 times the matched Architecture-A OL+BDC loss (2.8%). It falls below the 40-60% relative-loss range suggested by earlier exploratory runs in this project for the MVN differential-correlation DGP, but it confirms the qualitative finding of those runs, namely a substantial and strongly design-dependent erosion. We compare the carryover-induced power loss between architectures using a difference-of-differences z -statistic. Let $\Delta_X = \hat{\pi}_{X,0} - \hat{\pi}_{X,1}$ denote the within-architecture loss from $t_{1/2} = 0$ to $t_{1/2} = 1.0$ for architecture $X \in \{A, B\}$, estimated from $n = 1,000$ independent replicates at each cell. The B-versus-A gap is $\Delta_B - \Delta_A$, with standard error

$$SE = \sqrt{\frac{\hat{\pi}_{B,0}(1 - \hat{\pi}_{B,0})}{n} + \frac{\hat{\pi}_{B,1}(1 - \hat{\pi}_{B,1})}{n} + \frac{\hat{\pi}_{A,0}(1 - \hat{\pi}_{A,0})}{n} + \frac{\hat{\pi}_{A,1}(1 - \hat{\pi}_{A,1})}{n}}$$

since the four proportions come from independent replicate sets. All tests are two-sided. At OL+BDC this gives $z = 7.64$ ($p < 10^{-13}$); at Hybrid, $z = 3.96$ ($p < 10^{-4}$); the CO contrast is not statistically distinguishable from zero ($z = 0.29$) because the CO design's within-subject AR(1) structure dominates the carryover-mediated correlation decay channel that Architecture B uses to encode the biomarker-treatment interaction.

4.2 3.2 Why the architectures diverge

We turn now to a mechanistic account of why the two architectures produce such different power losses under carryover.

[16]

Architecture A is subject to only one carryover channel (mean blurring); it escapes the second channel (correlation erosion).

- Carryover compresses Dbc contrast and inflates off-drug BR means; both architectures share this channel.
- Architecture A lacks a second channel because β_{bm} is a fixed population parameter and the noise structure is drug-state-independent.
- Result: reduced variance in the interaction term, but signal-to-noise ratio degrades only modestly.

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Under Architecture A, carryover reduces the magnitude of the drug exposure variable seen by the analysis model (Dbc drops below 1 during the off-drug period), and the mean BR at off-drug timepoints is inflated by the residual drug effect exactly as in Architecture B. Both mechanisms compress the between-state treatment contrast and are shared by the two architectures. What Architecture A lacks is the second channel of signal loss: the biomarker moderation coefficient β_{bm} is a fixed population parameter, the biomarker enters the mean structure directly rather than through a correlation, and the noise structure is independent of drug state. The biomarker-by-drug interaction term

therefore has reduced variance (because Dbc has less contrast between drug states), but the signal-to-noise ratio degrades only modestly.

Under Architecture B, carryover operates on two levels simultaneously:

1. **Mean blurring:** The BR means during off-drug periods are inflated by residual carryover, making them resemble on-drug BR means. This reduces the treatment contrast that the analysis model uses to distinguish drug effect from no-drug. This channel is shared with Architecture A.
2. **Correlation erosion:** The BM-BR correlation during off-drug periods decays exponentially with t_{sd} . This is not just a statistical modeling choice; it is a structural consequence of the DGP. The correlation between biomarker and BR reflects the degree to which the drug effect is active; as the drug washes out, the correlation weakens because the drug-mediated component of BR variance shrinks relative to the drug-independent component. This channel is *unique* to Architecture B, because only here does the biomarker signal live in the second moment of the joint distribution. Under Architecture A the drug-mediated and drug-independent components of BR variance move in tandem (both are scaled by the same Dbc-weighted drug effect), so their ratio, and hence the identifiability of the interaction, is preserved even as amplitudes shrink.

[17]

Under Architecture B the interaction estimate is attacked from both sides simultaneously; Architecture A faces only one of the two attacks.

- The `bm:Dbc` coefficient depends on the covariance between B_i and Dbc-weighted outcomes.
- Under Architecture B: (1) Dbc contrast is compressed; (2) the BM-BR correlation generating the covariance signal is simultaneously decaying.
- Architecture A faces only attack (1), not attack (2); this single difference accounts for the divergence in power.

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The analysis model's interaction coefficient (the coefficient on `bm:Dbc`) depends on the covariance between B_i and Dbc-weighted outcomes. Under Architecture B, this covariance is being attacked from both sides: the treatment contrast in Dbc is compressed, and the BM-BR correlation that generates the covariance signal is simultaneously decaying. That Architecture A escapes the second attack entirely ac-

723 counts for the divergence in power loss documented above.

724 4.3 3.3 Robustness check at $N = 70$

725 [18]

726 **An $N = 70$ confirmation run checks whether the qualitative**
727 **ordering holds when power is near the ceiling.**

- 728 • 800 replicates per cell, 12 cells (2 architectures $\times 2$ c_{bm} $\times 3$ $t_{1/2}$),
729 OL+BDC design.
- 730 • 100% convergence across 9,600 fits.
- 731 • Purpose: verify the Section 3.1 ordering survives saturation at
732 the larger N .

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739 The Section~3.1 production results are at $N = 35$, where neither
740 architecture saturates power and the carryover-induced loss is me-
741 chanically visible. We conducted a confirmation run at $N = 70$ to
742 check whether the qualitative ordering survives the saturation that
743 comes with the larger sample size. An 800-replicate-per-cell Monte
744 Carlo run was executed at $N = 70$ across 12 cells (architecture
745 $\in \{\text{MVN, mean moderation}\} \times c_{bm} \in \{0, 0.45\} \times t_{1/2} \in \{0, 0.5, 1.0\}$
746 weeks), all on the OL+BDC design, using 8-core `parallel::mclapply`.
747 Wall-clock 483~s; convergence 100% across 9{,}600 fits.

748 At $c_{bm} = 0.45$:

Architecture	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
MVN (B)	0.978	0.945	0.885
Mean moderation (A)	0.976	0.973	0.959

749 [19]

750 **The $N = 70$ run confirms the qualitative ordering; compressed**
751 **absolute magnitudes reflect ceiling saturation, not a smaller**
752 **underlying effect.**

- 753 • Losses: 9.3 pp (Architecture B, $p < 10^{-12}$) vs. 1.7 pp (Architec-
754 ture A, $p \approx 0.05$); ordering matches $N = 35$.
- 755 • At $N = 70$ both architectures are within 2–5 pp of the ceiling at
756 $t_{1/2} = 0$, mechanically truncating observable loss.
- 757 • $N = 35$ production results remain the canonical reference for the
758 narrative magnitude claims.

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765 Within-architecture losses from $t_{1/2} = 0$ to $t_{1/2} = 1$ are 9.3 per-
766 centage points (Architecture~B; two-proportion z -test on the paired
767 cells, $z = 7.4$, $p < 10^{-12}$) and 1.7 percentage points (Architec-
768 ture~A; $z = 2.0$, $p \approx 0.05$), where each z -statistic uses the formula
769 $(\hat{\pi}_0 - \hat{\pi}_1) / \sqrt{\hat{\pi}_0(1 - \hat{\pi}_0)/n + \hat{\pi}_1(1 - \hat{\pi}_1)/n}$ with $n = 800$ replicates
770 per cell. The qualitative ordering matches Section~3.1's $N = 35$
771 result. The absolute magnitudes are compressed because $N = 70$
772 at the prazosin-PTSD calibration places both architectures within
773 2-5 percentage points of the unit ceiling at $t_{1/2} = 0$, mechanically
774 truncating the loss either architecture's curve can absorb. We note
775 that reading the narrative loss-magnitude claim against the $N = 70$
776 robustness numbers (3% and 9% relative loss for Architecture~B
777 versus 0.4% and 1.8% for Architecture~A) requires acknowledging
778 this saturation effect; the $N = 35$ production results in Section~3.1
779 are the canonical reference for the narrative magnitude.

780 [20]

781 **Type I error is well-controlled across all 18 null cells, with**
782 **no architecture-specific inflation.**

- 783 • Type I error range [0.010, 0.074], mean 0.043, median 0.042 across
784 18 null cells at $N = 35$.
- 785 • No architecture-specific inflation detected.
- 786 • The slightly conservative Architecture B OL+BDC null cell at
787 $t_{1/2} = 1.0$ ($p = 0.010$) is the only notable departure.

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794 **Type I error.** Across all 18 null cells of the $N = 35$ production
795 run, Type~I error sat in [0.010, 0.074], with mean 0.043 and median
796 0.042. We observe no architecture-specific inflation. The uniformly
797 sub-nominal rates are consistent with the corCAR1 working-covariance
798 miscalibration characterized in the companion calibration study¹;
799 because it affects both architectures alike, it is common-mode with
800 respect to the architecture contrast. Architecture~B's $t_{1/2} = 1.0$
801 OL+BDC null ($p = 0.010$, slightly conservative) is the smallest cell;
802 the corresponding alternative ($p = 0.539$) is the most informative cell

803 of the Section~3.1 narrative.

804 [21]

805 **Estimator bias is small and architecturally interpretable;**
806 **coefficient drift under Architecture A contrasts with**
807 **correlation-channel erosion under Architecture B.**

- 808 • Mean $\hat{\beta}_{bm:D_{bc}}$ at $c_{bm} = 0.45$ ranges -0.231 to -0.281 across
809 alternative cells; Architecture A OL+BDC at $t_{1/2} = 1.0$ produces
810 the largest absolute mean.
- 811 • Architecture B cells maintain mean near -0.234 across all $t_{1/2}$
812 values, consistent with erosion through correlation decay rather
813 than coefficient drift.
- 814 • Nominal-95% Wald coverage holds across the grid.

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821 **Estimator bias.** Mean estimated $\hat{\beta}_{bm:D_{bc}}$ at $c_{bm} = 0.45$ ranged from
822 -0.231 to -0.281 across the 18 alternative cells, with the $N = 35$
823 Architecture-A OL+BDC cell at $t_{1/2} = 1.0$ producing the largest ab-
824 solute mean (-0.281); the Architecture-B cells maintained mean near
825 -0.234 across all $t_{1/2}$ values, consistent with the Architecture-B infor-
826 mation channel being eroded through correlation decay rather than
827 through coefficient drift. Empirical SE of $\hat{\beta}_{bm:D_{bc}}$ runs 0.05 to 0.10
828 across cells; the within-replicate model SE matches at scale, indicat-
829 ing nominal-95% Wald coverage holds across the grid.

830

831 5 4. Biological Assumptions

832 [22]

833 **Architecture choice encodes a biological claim about how the**
834 **biomarker moderates treatment response.**

- 835 • The three architectures reflect distinct assumptions about the
836 mechanism of biomarker-treatment moderation.
- 837 • Section 4 takes each architecture in turn (4.1, 4.2, 4.3 for Archi-
838 tectures A, B, and C respectively), then discusses the prazosin-
839 PTSD motivating case.

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846 The choice between architectures is not merely a statistical conven-
847 nience; it reflects different assumptions about the biological mech-
848 anism by which the biomarker moderates treatment response. We
849 consider each architecture in turn, then take up the prazosin-PTSD
850 motivating application in detail.

851 5.1 4.1 When Architecture A is appropriate

852 [23]

853 **Architecture A is appropriate when the biomarker directly**
854 **determines the magnitude of the drug effect; the scaling is**
855 **deterministic at the mean-structure level.**

- 856 • The drug effect scales with the biomarker value: $\text{effect}_i = f(B_i) \cdot$
857 drug.
- 858 • 'Deterministic' means any two participants with the same B_i have
859 the same expected response; individual outcomes still carry noise.
- 860 • Appropriate when the biomarker is a direct pharmacoki-
861 netic/pharmacodynamic (PK/PD) determinant (eGFR, receptor
862 density, metabolizer status).

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869 Architecture A (direct mean moderation) is appropriate when the
870 biomarker governs the **magnitude of the drug's biological ef-**
871 **fect** for each individual participant. The drug effect for participant
872 i is scaled by their biomarker value: $\text{effect}_i = f(B_i) \cdot \text{drug}$. The scal-
873 ing is deterministic in the population-mean sense, in that any two
874 participants sharing the same B_i have the same expected response,
875 but individual outcomes still carry the usual noise, random-effect, and
876 residual-biological variability. The label 'deterministic' refers to the
877 mean structure of the DGP, not to the realized response of any single
878 participant.

879 Clinical examples:

- 880 • **Renal clearance of a renally-excreted drug.** Estimated glomerular
881 filtration rate (eGFR) directly governs the elimination rate of drugs such

as digoxin or aminoglycoside antibiotics. Participants with higher eGFR clear the drug more rapidly, achieving lower steady-state concentrations and a proportionally smaller drug effect. The biomarker-drug relationship is mechanistic and deterministic: given the biomarker value, the effective dose at steady state is fully determined (up to random noise).

- **Receptor density moderation.** A biomarker that measures target receptor density (e.g., PET imaging of receptor availability for an antagonist) directly determines the pharmacodynamic response to that antagonist. More receptors means more drug effect, proportionally.
- **Genetic metabolizer status.** CYP2D6 metabolizer phenotype determines the rate of drug activation or clearance. Poor metabolizers of a prodrug receive less active compound, reducing their treatment effect proportionally.
- **Dose-response with biomarker-dependent effective dose.** When the biomarker determines the effective dose (through body composition, protein binding, or volume of distribution), the treatment effect scales with the biomarker through the dose-response curve.

[24]

In all Architecture A cases, the proportional biomarker-drug relationship is preserved during the off-drug period.

- Residual drug effect during the off-drug period maintains the same proportional relationship with the biomarker.
- If participant A has $2\times$ the drug effect of B on-drug, they also have $\approx 2\times$ the residual effect during the off-drug period.
- This preserved proportionality is what Architecture A encodes and why it loses little power under carryover.

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In all these cases the biomarker's moderating effect is preserved under carryover because the residual drug effect during off-drug periods maintains the same proportional relationship with the biomarker. If participant A has twice the drug effect of participant B when on drug, they will also have approximately twice the residual effect during the off-drug period. That proportionality is precisely what Architecture A encodes.

5.2 4.2 When Architecture B is appropriate

[25]

923 **Architecture B is appropriate when the biomarker is a prob-**
924 **abilistic predictor of response, mediated by latent factors the**
925 **biomarker imperfectly indexes.**

- 926 • Appropriate when the biomarker predicts which participants are
927 likely to respond, not the magnitude of any individual's response.
- 928 • The biomarker-response association is population-level and prob-
929 abilistic, mediated by latent factors (disease subtype, neurobio-
930 logical heterogeneity, comorbidities).
- 931 • Two participants with identical biomarker values may respond
932 very differently.

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939 Architecture B (differential correlation) is appropriate when the
940 biomarker predicts **which participants are likely to respond**, but
941 the magnitude of any individual's response is not deterministically
942 governed by the biomarker. The biomarker and the drug response
943 are associated at the population level, but the association is mediated
944 by latent factors (disease subtype, neurobiological heterogeneity,
945 comorbidities) that the biomarker imperfectly indexes.

946 **Clinical examples:**

- 947 • **Blood pressure as a PTSD subtype marker.** Elevated resting blood
948 pressure may index a noradrenergic-predominant PTSD phenotype¹³ that is
949 more likely to respond to prazosin (an alpha-1 adrenergic antagonist). The
950 blood pressure does not *cause* the drug to work better; rather, it correlates
951 with an underlying neurobiological state that determines drug responsive-
952 ness. Two participants with identical blood pressure may have very different
953 responses because blood pressure is an imperfect proxy for the noradrenergic
954 state.
- 955 • **Baseline disease severity as a treatment predictor.** Trials in
956 Alzheimer's disease, depression, and chronic pain often find that patients
957 with more severe baseline symptoms show larger treatment effects (the
958 'floor effect' or 'regression to the mean' phenomenon). The baseline severity
959 score correlates with treatment response, but the relationship is statistical
960 (population-level tendency), not deterministic.
- 961 • **Inflammatory biomarkers in psychiatry.** C-reactive protein (CRP) or
962 interleukin-6 levels may predict response to anti-inflammatory augmentation
963 of antidepressants¹⁸. High-CRP patients are more likely to respond, but the
964 relationship is probabilistic: many high-CRP patients do not respond, and
965 some low-CRP patients do.

- 966 • **Genetic risk scores.** Polygenic risk scores predict disease trajectory and
967 treatment response, but explain only a fraction of the variance. The remain-
968 der reflects unmeasured genetic, epigenetic, and environmental factors.

969 [26]

970 **Under Architecture B, carryover weakens the biomarker-**
971 **response correlation because the biomarker's predictive**
972 **power depends on the presence of active drug.**

- 973 • As the drug washes out, the drug-mediated variance component
974 diminishes, weakening the BM-BR correlation.
- 975 • The biomarker's predictive power is tied to active drug effect;
976 without drug, it predicts nothing about non-drug symptom
977 change.
- 978 • This is the 'correlation erosion' channel unique to Architecture
979 B.

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986 In these cases carryover has a different implication: as the drug washes
987 out, the correlation between biomarker and response weakens because
988 the drug-mediated component of response variance diminishes. The
989 biomarker's predictive power is inherently tied to the presence of active
990 drug effect. Without drug, the biomarker is just a biomarker; it does
991 not predict the non-drug component of symptom change.

992 [27]

993 **Architecture B is a tractable second-moment approximation**
994 **to the biologically faithful finite-mixture DGP; adequate for**
995 **power calculations but not a mechanistic endorsement.**

- 996 • If the biomarker indexes a discrete responder class, the faith-
997 ful DGP is a finite mixture; Architecture B approximates its
998 population-level covariance structure.
- 999 • Architecture B captures the covariance implications of latent-
1000 class membership without committing to the full generative form.
- 1001 • Adequate for power calculations; should not be read as endorsing
1002 MVN differential correlation as the correct mechanism.

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The latent-subtype framing raises a legitimate modeling question: if the biomarker is really a noisy indicator of an underlying responder class, the biologically faithful DGP would be an explicit finite mixture in which each participant is drawn from one of two (or more) response distributions with class-membership probability a function of B_i . Under such a mixture DGP the individual-level moderation is discrete, not probabilistic, and the MVN differential-correlation parameterization is best understood as a tractable second-moment approximation to a richer mixture model rather than as the generative mechanism itself. The practical consequence is that Architecture~B as implemented here captures the population-level covariance implications of a latent-class mechanism without committing to its full generative form; this is adequate for power calculations that test for the presence of an interaction but should not be read as an endorsement of MVN differential correlation as the correct mechanistic model. Whether the observed power loss under carryover is better mitigated by analysis strategies (Section 6) or by explicit mixture modeling is taken up in a companion paper¹⁴, which derives the conditions under which a single multivariate normal reproduces the first two moments of a two-component mixture, catalogues the regimes in which that approximation fails, and finds in a preliminary evaluation that at trial-relevant sample sizes class-aware analyses are not competitive with the continuous linear-mixed baseline.

[28]

Senn (2016) cautions that apparent personalized response is often a variance artifact; Architecture B presupposes a real, stable latent subtype, which may not hold.

- Senn (2016) argues that apparent biomarker-treatment interactions are often variance-component artifacts, not genuine effects.
- Architecture B presupposes the latent subtype is real, identifiable, and stable across the trial period.
- If that presupposition is shaky, the documented power loss may overstate recoverable signal.

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The latent-subtype framing of Architecture B should also be adopted with appropriate epistemic caution. Senn¹⁹ has argued that the apparent ‘personal element in response to treatment’ is often a variance-

component artifact rather than a genuine biomarker-treatment interaction, and that even modest within-subject variability can produce convincing-looking but spurious differential response. Architecture B presupposes that the latent subtype indexed by the biomarker is real, identifiable, and stable across the trial period. When that presupposition is shaky, the apparent power loss under carryover documented in Section 3 may overstate what could be recovered under any analysis model, because the underlying signal it is attempting to estimate may itself be smaller than assumed.

5.3 4.3 The prazosin-PTSD case

We turn briefly to the motivating clinical application for pmsimstats, which is the active-duty prazosin trial by Raskind et al.¹⁵, whose SBP-stratified secondary analysis is the subject of ongoing reanalysis by the pmsimstats team. The biomarker is resting systolic blood pressure (SBP) and the outcome is CAPS (Clinician-Administered PTSD Scale) score change.

[29]

The prazosin-PTSD evidence base is mixed; the PACT null result motivates the search for a predictive biomarker, with SBP as the leading candidate.

- Earlier single-site Raskind trials (2003, 2007, 2013) reported benefit on nightmares and PTSD symptoms.
- PACT (2018) failed to replicate; this has been interpreted as consistent with effect heterogeneity.
- SBP, as a noninvasive proxy for central noradrenergic tone, is the leading predictive biomarker candidate.

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The clinical evidence base for prazosin in PTSD is mixed. The earlier single-site trials by Raskind and colleagues^{15,20,21} reported benefit on nightmares and global PTSD symptoms in veteran and active-duty samples. The subsequent multisite Veterans Affairs Cooperative Study Program trial¹⁶, known as PACT, did not replicate the effect: prazosin failed to outperform placebo on the primary endpoints. The negative PACT result has been interpreted by several investigators as consistent with effect heterogeneity, namely that prazosin works in some patients but not others, which motivates the search for a predictive biomarker that would identify the responder subgroup. Resting SBP, as a noninvasive proxy for central noradrenergic tone, is the leading

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candidate.

[30]

The SBP-prazosin mechanism is an Architecture B scenario: SBP is a proxy for noradrenergic state, not a direct PK determinant.

- Elevated SBP indexes heightened noradrenergic tone, a PTSD subtype preferentially responsive to prazosin.
- SBP is a proxy, not a causal determinant; two patients with identical SBP may have very different noradrenergic states.
- This is precisely the Architecture B scenario: probabilistic association, not deterministic scaling.

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The biological hypothesis is that elevated SBP indexes heightened central noradrenergic tone, which characterizes a subtype of PTSD that is preferentially responsive to alpha-1 adrenergic blockade by prazosin. This is an Architecture B scenario: SBP is a proxy for an underlying neurobiological state, not a direct determinant of drug pharmacokinetics. Two patients with the same SBP may differ in their actual noradrenergic tone, alpha-1 receptor distribution, and downstream signaling, leading to different treatment responses despite identical biomarker values.

[31]

Hendrickson et al. (2020) correctly uses Architecture B for this context; the carryover power loss is clinically relevant for trial design decisions.

- Architecture B (MVN differential correlation) is appropriate for the SBP-prazosin mechanism.
- The documented carryover sensitivity means designs with longer off-drug periods will have meaningfully less power.
- This effect is larger than a direct mean moderation simulation would predict.

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The Hendrickson et al.¹³ DGP uses Architecture B (MVN differential correlation) for this clinical context. The finding that carryover substantially reduces power under this architecture is therefore clinically relevant: it suggests that trial designs with longer off-drug periods will have meaningfully less power to detect the SBP-prazosin interaction than designs with shorter off-drug periods, and this effect is larger than what a direct mean moderation simulation would predict.

1142

6 5. Implications for Simulation Study Design

6.1 5.1 Choosing the appropriate architecture

The choice of DGP architecture should be guided by the biological mechanism hypothesized for the biomarker-treatment interaction:

Criterion	Architecture A	Architecture B
Biomarker role	Causal mediator	Statistical predictor
Mechanism	Deterministic scaling	Probabilistic association
Signal location	Mean structure	Covariance structure
Free parameters	1 (c_{bm})	1 (c_{bm})
Carryover impact	Modest (1–3%)	Design-dependent, up to 31%
Appropriate when	Biomarker determines dose or PK	Biomarker indexes a latent subtype

When the operative mechanism is unknown, Architecture-B is the conservative default for trial-design decisions: its larger carryover sensitivity keeps power estimates from being optimistic relative to Architecture-A. A combined architecture that mixes the two channels, together with a corresponding sensitivity sweep, is developed in the companion paper¹⁷.

6.2 5.2 Reporting requirements

Simulation studies evaluating biomarker-moderated treatment effects should explicitly state:

1. Whether the biomarker-treatment interaction is generated through mean moderation, differential correlation, or both channels simultaneously.
2. The biological rationale for the chosen mechanism.
3. Whether the carryover model (if present) operates on the interaction signal consistently with the chosen architecture.
4. Sensitivity analyses under alternative architectures, if the biological mechanism is uncertain.

[32]

The four reporting requirements are the ADEMP 'D' element applied specifically to biomarker-validation simulations.

- ADEMP (2019) prescribes five elements: Aims, Data-generating mechanisms, Estimands, Methods, Performance measures.
- The DGP-architecture choice concerns the D element; the four requirements above specialize that element for architecture-sensitive simulations.

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The reporting framework above is consistent with the broader ADEMP template of Morris, White, and Crowther²², which prescribes that simulation studies fix five elements before production runs: the **Aims** of the study, the **Data-generating mechanisms**, the **Estimands** (the inferential targets), the **Methods** (analysis specifications compared), and the **Performance measures** (power, bias, coverage, Type I error, and MCSEs of each). The DGP-architecture choice this paper concerns is the **D** element of an ADEMP specification; the reporting requirements above are the specialization of the broader ADEMP framework to the architecture-sensitive class of biomarker-validation simulations.

[33]

Primary estimand: $\beta_{bm:D}$ in the linear mixed-effects (LME) model; performance measures cover power, Type I error, bias, SE, and convergence; 1000 replicates per cell.

- Estimand: $\beta_{bm:D}$ in $Sx \sim bm + t + Dbc + bm:Dbc$ with random intercept and AR(1).
- Five performance measures: power, Type I error, bias, empirical SE, convergence rate.
- $n_{\text{reps}} = 1000$ throughout; calibrated to achieve $\text{MCSE} < 0.02$ for power near 0.5.

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The **primary estimand** for this paper's simulations is the biomarker-

treatment interaction coefficient $\beta_{bm:D}$ in the linear-mixed analysis model $Sx \sim bm + t + Dbc + bm:Dbc$ with random intercept and AR(1) residual correlation. The reported **performance measures** are: (i) power for $\beta_{bm:D}$ at $\alpha = 0.05$, with MCSE $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$; (ii) Type I error under $c_{bm} = 0$, with the same MCSE; (iii) bias of $\hat{\beta}_{bm:D}$ relative to the calibrated true value at each parameter cell; (iv) the empirical MCSE of $\hat{\beta}_{bm:D}$ across replicates; and (v) the convergence rate of the linear-mixed fit. Sample size, biomarker effect size, and carryover half-life vary across the parameter grid; the number of replicates per cell is calibrated to achieve MCSE below 0.02 for power at $\hat{\pi} = 0.5$, which corresponds to $n_{\text{reps}} \geq 625$. The simulations reported here use $n_{\text{reps}} = 1000$ throughout.

6.3 5.3 When the choice is uncertain

If the biological mechanism is ambiguous, as it often is in early biomarker development, we suggest investigators take the following steps:

1. Run the simulation under both architectures and report the range of power estimates.
2. Architecture B produces more conservative (lower) power estimates under carryover, making it the safer default for trial design decisions.
3. Design the trial to minimize carryover (adequate washout periods) to reduce sensitivity to the DGP architecture choice.

7 6. Alternative Analysis Strategies Under Architecture B

This section provides a conceptual overview of six candidate strategies for mitigating the power loss documented in Section 3. None of the strategies below has been evaluated by simulation within this study; the assessments of expected benefit are qualitative and based on the mechanistic account in Section 3.2. A systematic simulation comparison is warranted before any strategy is recommended for applied use.

The power loss documented in Section 3 arises from two distinct sources: (1) compressed variance of the drug exposure variable Dbc when carryover is present, and (2) erosion of the BM-BR correlation during off-drug periods. Source 1 is a property of the analysis model parameterization and can potentially be addressed by choosing a different predictor. Source 2 is a property of the data itself (the signal is genuinely weaker in off-drug observations), and no analysis model can recover signal that is not present. However, analysis strategies that are selective

1241 about *which observations contribute to the test* or *how they are weighted* can
1242 mitigate the damage.

1243 7.1 6.1 Restrict analysis to on-drug observations

1244 The most direct approach is to discard off-drug timepoints entirely and test the
1245 biomarker-treatment interaction using only on-drug observations. Every retained
1246 observation has the full BM-BR correlation (c_{bm}), so correlation erosion is elim-
1247 inated. The costs are twofold: a reduction in effective sample size (fewer obser-
1248 vations per participant) and, more importantly, the loss of the off-drug reference
1249 that separates the biological response from placebo and time-varying background
1250 response. Without off-drug observations the biological, placebo, and time-varying
1251 components are confounded, which is why the Open-Label design, despite having
1252 the most on-drug timepoints, does not dominate the other designs in the Section 3
1253 results. For designs with many on-drug timepoints (OL has 8, the Hybrid design
1254 has 5-6 per path), the net effect may be positive: the per-observation signal-to-
1255 noise ratio is maximal, and the analysis model simplifies to $\mathbf{Sx} \sim \mathbf{bm} + \mathbf{t} + \mathbf{bm}:\mathbf{t}$
1256 since there is no treatment contrast to model. The interaction is detected through
1257 the correlation between biomarker and the time-on-drug trajectory.

1258 [34]

1259 **Restricting to on-drug observations eliminates correlation**
1260 **erosion but loses the off-drug reference; best for OL and Hy-**
1261 **brid designs, worst for CO.**

- 1262 • Benefit: every retained observation has full c_{bm} ; correlation ero-
1263 sion is eliminated.
- 1264 • Cost: loss of off-drug reference confounds biological, placebo, and
1265 time-varying components.
- 1266 • Practical: best for OL/Hybrid (many on-drug timepoints); least
1267 useful for CO (discards half the observations).

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1274 This approach is most promising for the Open-Label design (which
1275 has no off-drug observations regardless) and the Hybrid design (which
1276 has a long initial on-drug phase). It is least useful for the Crossover
1277 design, where discarding half the observations substantially reduces
1278 statistical power.

1279 7.2 6.2 Weighted analysis

1280 [35]

Weighted analysis downweights off-drug observations by drug exposure without discarding them; information-theoretic rationale; straightforward to implement.

- Weight: $w_t = e^{-\lambda t_{sd}}$ for off-drug observations; 1 for on-drug.
- Rationale: higher exposure = stronger correlation signal = more information about the interaction.
- Implemented via the `weights` argument in `nlme::lme`.

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Rather than discarding off-drug observations entirely, a weighted analysis retains all observations but weights them by their estimated drug exposure level. On-drug observations receive weight 1. Off-drug observations receive weight proportional to the expected residual drug effect:

$$w_t = e^{-\lambda \cdot t_{sd}}$$

This downweights observations where the BM-BR correlation has decayed (and where they contribute more noise than signal to the interaction estimate) without discarding them entirely. The rationale is information-theoretic: observations with higher drug exposure carry more information about the biomarker-treatment interaction because the correlation signal is stronger. This is straightforward to implement in `nlme::lme` using the `weights` argument.

6.3 Within-subject contrast

For designs with both on-drug and off-drug periods (CO, Hybrid, OL+BDC), a summary-measure approach computes a per-participant contrast: the difference between mean on-drug response and mean off-drug response. The biomarker-treatment interaction is then tested by a simple regression of these contrasts on the biomarker:

$$\bar{Y}_{i,\text{on}} - \bar{Y}_{i,\text{off}} = \alpha + \gamma \cdot B_i + \epsilon_i$$

[36]

Within-subject contrast sidesteps the repeated-measures model; trades efficiency for robustness to carryover misspecification.

- Per-participant contrast $\bar{Y}_{\text{on}} - \bar{Y}_{\text{off}}$ is regressed on the biomarker.

- Carryover compresses the contrast magnitude but the contrast-biomarker correlation may be more robust.
- Trade-off: less efficient than a full longitudinal model, but more robust to carryover and correlation misspecification.

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This sidesteps the Dbc parameterization and the repeated-measures model entirely. Carryover affects the magnitude of the contrast (making off-drug means closer to on-drug means), but the *correlation between the contrast and the biomarker* may be more robust because within-subject averaging reduces noise before the biomarker association is tested. The approach trades the efficiency of a full longitudinal model for robustness to misspecification of the carryover and correlation structure.

7.4 6.4 Two-stage random slopes

A two-stage approach first estimates participant-specific treatment effects, then tests their association with the biomarker. Stage 1 fits a random-slopes model:

$$Y_{it} = \beta_0 + \beta_1 t + \beta_2 D_{it} + u_{0i} + u_{1i} D_{it} + \epsilon_{it}$$

where β_2 is the population mean drug effect and u_{1i} is participant i 's random deviation from that mean. The participant-specific drug effect is then $\beta_2 + u_{1i}$. Stage 2 regresses the estimated random deviations (or equivalently, the best linear unbiased predictors (BLUPs) of the participant-specific effects) on the biomarker:

$$\hat{u}_{1i} = \delta_0 + \delta_1 B_i + \eta_i$$

[37]

Two-stage random slopes separates within-subject effect estimation from between-subject biomarker association; may be more robust to carryover.

- Stage 1: random-slopes LME estimates participant-specific drug effects ($\beta_2 + u_{1i}$).
- Stage 2: regress BLUPs on biomarker to test association.
- The random slope u_{1i} integrates over the full trajectory; robustness to carryover is plausible but unverified by simulation.

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This separates the within-subject treatment effect estimation (which uses the longitudinal data and is affected by carryover) from the between-subject biomarker association (which uses the cross-sectional relationship between estimated effects and biomarker values). The approach may be more robust to carryover because the random slope u_{1i} integrates over the participant's entire trajectory, including both on-drug and off-drug phases.

7.5 6.5 Exclusion of contaminated observations

[38]

Excluding the first 1–2 off-drug timepoints removes the most contaminated observations; later off-drug timepoints provide a cleaner contrast.

- Early off-drug timepoints: highest residual drug effect, BM-BR correlation actively decaying.
- Later off-drug timepoints: fully washed out, cleaner contrast with on-drug phase.
- Result: comparison between fully-on and fully-off states, avoiding the strained intermediate zone.

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The first 1-2 off-drug timepoints carry the most carryover contamination: the residual drug effect is highest, and the BM-BR correlation is in the process of decaying. Later off-drug timepoints, where the drug has fully washed out, provide a cleaner contrast with the on-drug phase. Excluding the early off-drug observations and retaining only the later ones yields a comparison between fully-on and fully-off states, avoiding the intermediate zone where the analysis model is most strained.

7.6 6.6 Design-level solutions

Rather than adapting the analysis model to accommodate carryover, the trial design itself can be modified to reduce carryover exposure:

- **Longer washout periods** between drug phases eliminate residual drug

1393 effect before off-drug measurements begin.

1394 • **More on-drug timepoints** relative to off-drug increase the proportion of
1395 high-signal observations.

1396 • **Front-loaded drug exposure** (as in the OL+BDC design) places a
1397 long initial on-drug block before the off-drug baseline-during-continued-
1398 treatment phase, so that the BM-BR correlation has been fully established
1399 before any off-drug observations are collected. The alternative ordering
1400 (off-drug first, then on-drug) has the appealing property of placing the
1401 placebo and time-varying estimation in an uncontaminated window but
1402 sacrifices on-drug exposure duration and forces the BM-BR signal to be
1403 re-established after a wash-in; the net power trade-off depends on the ratio
1404 of carryover half-life to total trial length and is explored empirically in the
1405 Section 3 results.

1406 [39]

1407 **Design-level changes address the root cause rather than the**
1408 **symptom; cost is longer trial or reduced ability to estimate**
1409 **carryover dynamics.**

- 1410 • Three levers: longer washout, more on-drug timepoints, front-
1411 loaded drug exposure (as in OL+BDC).
- 1412 • Front-loading establishes BM-BR correlation before any off-drug
1413 observations; alternative ordering (off-drug first) trades clean
1414 placebo estimation for reduced on-drug exposure.
- 1415 • Addresses root cause; cost is typically longer trial or reduced
1416 carryover estimation ability.

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1423 Design-level solutions address the root cause (carryover in the data)
1424 rather than the symptom (power loss in the analysis). Their cost is
1425 typically a longer trial duration or reduced ability to estimate the
1426 carryover dynamics themselves.

1427 7.7 6.7 Qualitative comparison

1428 Table~1 summarizes the strategies along three dimensions: whether they discard
1429 observations, implementation complexity, and the expected direction of benefit
1430 based on the mechanistic account in Section 3.2. All entries in the ‘Hypothesized
1431 benefit’ column are qualitative; no simulation evidence is presented here, and each
1432 strategy should be evaluated empirically before use in trial planning.

Table 1: Qualitative summary of analysis strategies for recovering power under Architecture B carryover. All benefit assessments are mechanistic hypotheses, not simulation-supported estimates.

Strategy	Discards data	Complexity	Hypothesized benefit
On-drug only	Yes	Low	Eliminates correlation erosion; forfeits off-drug reference (best for OL/Hybrid)
Weighted	No	Low	Reduces erosion damage proportionally; depends on weight calibration
Within-subject contrast	Aggregates	Low	Robust to carryover misspecification; less efficient than LME
Two-stage random slopes	No	Moderate	Unknown; separates estimation stages; robustness unverified
Exclude early off-drug	Some	Low	Removes most contaminated observations; improves contrast clarity
Design modification	N/A	N/A	Addresses root cause; cost is longer trial or reduced carry-over estimation

A systematic simulation comparison of these strategies under Architecture B, analogous to the factorial comparison of analysis models in Section 3, would identify which approach best recovers the power lost to carryover for each trial design. We regard this comparison as important future work and note that the on-drug-only restriction and the weighted analysis are the most tractable candidates for a first empirical evaluation.

7. Relationship to the Broader Literature

7.1 Treatment effect heterogeneity

[40]

The PATH Statement distinguishes risk modeling from effect modeling for predictive HTE analysis; this is an analysis-model distinction, not a DGP distinction.

- Rothwell (2005) establishes the predictive/prognostic biomarker distinction; Kent et al. (2020) synthesize it into the PATH Statement.

- PATH’s risk modeling approach stratifies on a multivariable risk score; effect modeling adds explicit interaction terms.
- PATH favors risk modeling and cautions against data-driven effect modeling because of low power, multiplicity, and overfitting.

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The statistical literature on treatment effect heterogeneity is extensive. Rothwell³ sets out the foundational distinction between *predictive* biomarkers (those that modify the treatment effect) and *prognostic* biomarkers (those that predict outcome regardless of treatment). The current consensus statement on predictive HTE analysis is the Predictive Approaches to Treatment effect Heterogeneity (PATH) Statement of Kent et al.⁴, which distinguishes two analytic strategies for identifying predictive biomarker effects in randomized trials: a *risk modeling* approach, in which a multivariable risk model (developed without a treatment term) is used to stratify patients and examine variation in absolute treatment benefit across risk strata, and an *effect modeling* approach, in which the regression includes explicit treatment-by-covariate interaction terms. PATH expresses caution about data-driven effect modeling on the grounds of low statistical power, multiplicity, and overfitting, and favors risk modeling when the application criteria are met.

[41]

PATH’s analysis-model axis (risk vs. effect modeling) is orthogonal to our DGP axis; both PATH strategies implicitly assume Architecture A.

- PATH’s distinction is analysis-model; ours is DGP; they are orthogonal.
- Both PATH strategies assume heterogeneity is in the mean structure (Architecture A); Architecture B does not fit either PATH category.
- Architecture A = parametric interaction model; Architecture B = latent-class/mixture-model perspective.

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We note that the PATH risk-versus-effect-modeling distinction is an analysis-model axis. The Architecture A versus Architecture B distinction introduced in this paper is a data-generating-process axis, and the two are orthogonal. Both PATH analytic strategies implicitly assume that the underlying treatment effect heterogeneity is encoded in the mean structure of the outcome (the heterogeneity is identifiable from differences in conditional means), which corresponds to Architecture A as a DGP. Architecture B as a DGP is a separate phenomenon that does not map cleanly onto either of PATH’s analysis-model categories: the interaction signal resides in the covariance structure rather than the mean, and the standard regression-based predictive HTE analysis methods endorsed by PATH may have reduced power under Architecture B precisely because they were developed under an implicit Architecture A assumption. Architecture A can accordingly be characterized as a *parametric interaction model* (in PATH’s effect modeling sense). Architecture B can be characterized as a *latent class* or *mixture model* perspective in which the biomarker is a noisy indicator of class membership.

7.2 Prevalence of each architecture

[42]

Architecture A is the near-universal standard across the biomarker-moderated trial simulation literature; Architecture B is absent from all major threads except Hendrickson et al.

- Simon (2010), Freidlin and Korn (2014), Renfro and Sargent (2017), Haller and Ulm (2018), and the platform trial literature all use Architecture A.
- N-of-1 simulation studies (1997, 2016, 2013) use hierarchical random-effects models, a variant of Architecture A.
- Wang and Schork (2019) and Schork (2022) share authorship with Hendrickson et al. yet adopt Architecture A, suggesting the differential-correlation choice is paper-specific.

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A survey of the simulation methodology literature reveals that Architecture A (direct mean moderation) is the near-universal standard. Simon⁵ employs it throughout the simulation frameworks he develops for biomarker-stratified designs; Freidlin and Korn⁶ adopt it for

adaptive enrichment trials; Renfro and Sargent⁷ use it for basket and umbrella trials; and it underlies the platform trial literature throughout. Haller and Ulm⁸, in simulation work on biomarker-treatment interaction estimation in randomized trials with prognostic covariates, likewise adopt the standard regression specification, focusing on power and bias of the interaction estimator under different covariate adjustment strategies. N-of-1 simulation studies, including Zucker et al. (1997), Araujo et al.¹⁰, and Duan et al.²³, use hierarchical random-effects models in which individual treatment effects are drawn from a population distribution, a variant of Architecture A in which the biomarker predicts the random treatment effect through the mean structure. Methodological work on power and design for N-of-1 trials with serial correlation and carryover, from both Wang and Schork¹¹ and Schork¹², adopts the same Architecture A specification, even though this work shares senior authorship with Hendrickson et al.¹³. It appears, then, that the differential-correlation parameterization in the latter is a paper-specific design choice rather than a methodological convention of any laboratory or research program.

[43]

Architecture B appears in latent variable / structural equation modeling (SEM) / copula contexts but is unique in N-of-1 clinical trial simulation.

- Architecture B appears in latent variable / SEM / Bayesian joint modeling / copula contexts (e.g., Rizopoulos 2012).
- In N-of-1 clinical trial simulation, Hendrickson et al. (2020) appears unique in using it.

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Architecture B (differential correlation via MVN) appears primarily in latent variable and structural equation modeling contexts, including the joint modeling framework discussed by Rizopoulos²⁴ (where treatment-state-dependent covariance relates longitudinal biomarkers to survival outcomes rather than to treatment response directly), Bayesian joint modeling frameworks, and copula-based simulation studies. In the clinical trial simulation literature specifically, the Hendrickson et al. (2020) framework appears to be unique in using differential correlation as the mechanism for biomarker-treatment interaction in an N-of-1 trial context.

[44]

The near-universal use of Architecture A means most published power estimates are optimistic for Architecture B biomarkers under carryover.

- Virtually all published power estimates for biomarker-moderated trials assume Architecture A.
- These estimates may be optimistic for biomarkers that operate through Architecture B (covariance-structure mechanism).
- The additional carryover-induced power loss under Architecture B is not accounted for.

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This predominance of Architecture A has an important implication: the power estimates reported in the vast majority of biomarker-moderated trial simulation studies may be optimistic for biomarkers that operate through an Architecture B mechanism, because these studies do not account for the additional power loss that carryover produces when the interaction signal resides in the covariance structure rather than the mean structure.

8.3 7.3 Crossover and N-of-1 trial methodology

[45]

The crossover trial literature addresses carryover as a nuisance parameter but ignores the architectural distinction; the standard recommendation applies differently under each.

- Senn (2002), Jones and Kenward (2014), and Sturdevant and Lumley (2021) treat carryover as a nuisance parameter; none engages with DGP architecture.
- Standard recommendation (include carryover in model or use adequate washout) applies to both architectures.
- Under Architecture B, crossover designs with short washout periods may be less suitable than previously recognized.

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The crossover trial literature, represented most thoroughly by Senn²⁵ and Jones and Kenward²⁶, addresses carryover extensively but does

not distinguish between architectures for biomarker-treatment interaction modeling. More recent methodological work by Sturdevant and Lumley²⁷ on testing for carryover effects via mixed-effects models likewise treats carryover as a nuisance parameter to be estimated and adjusted for, without engaging with how carryover interacts with biomarker-treatment effect signals. The standard recommendation, namely to always include carryover in the analysis model or to use adequate washout, applies to both architectures but has different power implications under each. The finding that Architecture B is more sensitive to carryover suggests that crossover designs with short washout periods may be less suitable for detecting correlation-based biomarker interactions than previously recognized.

8.4 7.4 Precision medicine trial design

[46]

Enrichment and biomarker-stratified designs assume Architecture A for power calculations; the architectural choice matters differently across three design families.

- Simon (2010) and Freidlin and Korn (2014) frame power calculations in the Architecture A regression idiom.
- Three design families warrant separate consideration: adaptive enrichment, treatment-switching, and basket/umbrella trials.
- The architectural assumption is rarely made explicit, so the practical implications are easy to overlook.

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Enrichment and biomarker-stratified designs, as discussed by Simon⁵ and Freidlin and Korn⁶, use Architecture A exclusively for power calculations. Three families of design are particularly worth considering in light of the architectural distinction.

[47]

Adaptive enrichment trials are vulnerable to Architecture B power shortfalls precisely when the biomarker mechanism is most uncertain.

- Interim power calculations driving adaptive decisions use Architecture A; if Architecture B holds and off-drug periods occur, realized power will be lower than predicted.

- The mismatch biases adaptation rules toward premature futility stopping or overly narrow enrichment criteria.
- The problem is worst when the biomarker mechanism is unknown, exactly when adaptive enrichment is most often proposed.

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Adaptive enrichment trials adjust the eligible population mid-trial based on emerging evidence of a biomarker-treatment interaction. The interim power calculations that drive these adaptive decisions are universally framed in the Architecture A regression idiom. If the true biomarker mechanism is closer to Architecture B and any participants pass through washout intervals between dose-finding stages, the realized power for the interaction test will be lower than the adaptation rule expects, biasing decisions toward premature futility stopping or toward overly narrow enrichment criteria. The discrepancy is most acute when the biomarker mechanism is uncertain at the design stage, which is precisely the situation in which adaptive enrichment is most attractive.

[48]

Treatment-switching designs suffer Architecture B power loss through off-drug correlation erosion; parallel-group enrichment designs do not, because there is no off-drug phase.

- Crossover, basket, platform, and N-of-1 hybrid designs all cycle participants through drug states; each off-drug phase erodes BM-BR correlation under Architecture B.
- Parallel-group enrichment designs escape this loss because each participant stays on a single treatment throughout.
- The architectural choice matters most in within-subject designs, where it is also most likely to be clinically important.

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Treatment-switching designs (most crossover, basket, and platform trials, as well as N-of-1 hybrid designs) cycle individual participants through different drug states. Under Architecture B, each off-drug phase erodes the BM-BR correlation, so the biomarker-treatment

interaction signal in the second and subsequent treatment periods is structurally weaker than in the first. The power loss documented in Section 3 applies here directly. By contrast, parallel-group enrichment designs, in which each participant remains on a single treatment for the duration of the trial, escape this loss because there is no off-drug phase during which the correlation can decay; for these designs, Architectures A and B yield similar power estimates. The architectural choice therefore matters most in exactly the designs where the biomarker mechanism is most likely to be of clinical interest, namely those that deliberately expose each participant to both treatment states in order to estimate within-subject responsiveness.

[49]

Basket and umbrella trials face an additional risk of optimistic aggregated power estimates when Architecture B holds in even a subset of arms.

- Renfro and Sargent (2017) discuss power calculations that average across biomarker-defined arms; the calculation assumes Architecture A throughout.
- Mixed architectures across baskets produce optimistic trial-level power estimates that are difficult to detect from aggregate results.
- Subgroup-specific power diagnostics under both architectures are warranted when the biological mechanism is heterogeneous.

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Basket and umbrella trials, as discussed by Renfro and Sargent⁷, introduce an additional layer because their power calculations average across multiple biomarker-defined arms. If even a subset of the arms operate through an Architecture B mechanism while the trial-wide power calculation assumes Architecture A throughout, the aggregated power estimates will be biased optimistically in a way that is difficult to detect from trial-level results alone. Subgroup-specific power diagnostics under both architectures should accompany the primary planning analysis whenever the underlying biology is heterogeneous across baskets.

[50]

Across all three design families, the practical recommendation is to anchor sample size planning on the Architecture B estimate when the biomarker mechanism is uncertain.

- Run power calculations under both architectures and treat Architecture B as the conservative anchor.
- This recommendation parallels the advice in Section 5.3 and is consistent with the PATH Statement (2020).
- The PATH Statement’s caution about data-driven interaction discovery reinforces the case for conservative planning under architectural uncertainty.

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In all three settings the practical recommendation parallels that of Section 5.3: when the biological mechanism is uncertain, run the power calculation under both architectures and treat the Architecture B estimate as the more conservative anchor for sample size planning. This recommendation is consistent with the PATH Statement⁴, which itself counsels caution about data-driven discovery of treatment-by-covariate interactions and prefers risk modeling approaches when the application criteria are met.

9 8. Conclusions

In conclusion, six points are to be emphasized.

1. The choice between direct mean moderation (Architecture A) and differential correlation (Architecture B) for generating biomarker-treatment interactions in clinical trial simulations has substantive consequences for power estimates under carryover.
2. Under Architecture A, carryover reduces power only modestly, by 1 to 3% in relative terms across the three designs examined, because the proportional biomarker-drug relationship is preserved during the off-drug period. Under Architecture B the loss is strongly design-dependent: it remains modest in the crossover and hybrid designs (3 to 5%) but reaches roughly 31% in the open-label OL+BDC design, because the differential-correlation signal erodes as the drug effect wanes.
3. The choice should be guided by the hypothesized biological mechanism: Architecture A for biomarkers that directly determine drug pharmacokinetics or pharmacodynamics; Architecture B for biomarkers that statistically predict treatment response through latent subtype membership.

4. For the prazosin-PTSD application with blood pressure as the predictive biomarker, Architecture B (MVN differential correlation) is the more appropriate model, and the resulting power sensitivity to carryover should inform trial design decisions regarding off-drug period duration.
5. The existing clinical trial simulation literature uses Architecture A almost exclusively. The Hendrickson et al. (2020) MVN approach is, to the best of our knowledge, unique in the N-of-1 trial context. This means that published power estimates for biomarker-moderated crossover and N-of-1 designs may be systematically optimistic for biomarkers that operate through a correlation-based (Architecture B) mechanism, because they do not account for the carryover sensitivity documented here.
6. Simulation studies for biomarker-moderated treatment effects should explicitly report their DGP architecture and consider sensitivity analyses under the alternative when the biological mechanism is uncertain.

10 Conflict of interest

The authors declare no conflicts of interest.

11 Funding

This work received no specific funding.

12 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

13 Reproducibility

This manuscript was rendered with R~4.6.0 inside the project's zzcollab Docker container (`rocker/tidyverse:4.6.0`; image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1`

residual correlation), `parallel` with `mclapply` (Section~3.1 and 3.3 production runs), `data.table` (the original-extended simulation collection), `purrr` (the `tidyverse` simulation collection), and `rmarkdown` for the manuscript build.

Sample-size convention. The Section~3.1 driver (`01-dgp-prototype.R`) generates $N = 35$ subjects independently per randomization path, then concatenates paths before analysis; for two-path designs (CO, OL+BDC) the effective analysis sample is therefore 70 subjects, and for the four-path Hybrid design it is 140 subjects.

Random-number generators. The Section~3.1 driver sets `set.seed(20260509)` and uses per-replicate seeds derived as `100000 * cell_index + rep_index`.

Data and code availability. Sections 3.1 and 3.3: per-replicate outputs at `analysis/data/quick-sim/01-dgp-replicates.rds`; ADEMP cell-level summaries at `01-dgp-summary.txt`; driver at `analysis/scripts/quick-sim/01-dgp-prototype.R`. The manuscript source, bibliography, and CSL file are at `analysis/report/01-dgp-mean-moderation-vs-mvn/`

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